



PRODUCT NAME

Industrial-grade Organic Shower Gel Formula

Description

A high-performance, environmentally conscious shower gel that offers gentle cleansing with biodegradable and moisturizing ingredients. This sulfate-free formulation is pH balanced and ideal for everyday use.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	12 months
Batch Size:	100 kg - 500 kg
Processing Time:	3 hours
Temperature:	40°C to 70°C during processing
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing vessels, agitator, heater, pH meter
Certification:	COSMOS Organic, Cruelty-Free Certified

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	45.0%	Solvent
2	Coco-Glucoside	Coco-Glucoside	15.0%	Mild surfactant, foaming agent
3	Decyl Glucoside	Decyl Glucoside	10.0%	Cleansing agent, biodegradable surfactant
4	Vegetable Glycerin	Glycerin	10.0%	Moisturizer, humectant
5	Xanthan Gum	Xanthan Gum	2.0%	Thickener, stabilizer
6	Citric Acid	Citric Acid	1.0%	pH adjuster
7	Caprylyl/Capryl Glucoside	Caprylyl/Capryl Glucoside	5.0%	Mild surfactant, foaming agent
8	Sodium Benzoate	Sodium Benzoate	1.0%	Preservative
9	Aloe Vera Leaf Juice	Aloe Barbadensis Leaf Juice	10.0%	Moisturizing agent
10	Panthenol	Panthenol	1.0%	Moisturizer, humectant
Total			100%	

Manufacturing Process

Phase 1: Phase A

1. Combine Aqua, Glycerin, and Xanthan Gum while stirring.
2. Heat mixture to 70°C maintaining constant stirring until Xanthan Gum is completely hydrated.

Phase 2: Phase B

1. Add Coco-Glucoside, Decyl Glucoside, and Caprylyl/Capryl Glucoside to the hydrated Phase A while stirring.

2. Maintain gentle agitation to avoid excessive foaming.

Phase 3: Phase C

1. Cool mixture to 40°C and add Aloe Barbadensis Leaf Juice and Panthenol.
2. Adjust pH using Citric Acid to reach 7.
3. Incorporate Sodium Benzoate as a preservative.

Phase 4: Phase D

1. Ensure batch is homogeneous and transfer to containers.
2. Label and package the product appropriately.

Required Equipment

Mixing vessels, agitator, heater, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: COSMOS Organic, Cruelty-Free Certified

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100 kg - 500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply an adequate amount of shower gel to damp skin. Gently lather and rinse thoroughly with water.